



Qwen-CUA: Native Computer Use for (almost) Everything

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Abstract

Computer-use agents must perceive interfaces, execute native actions, and sustain coherent behavior across diverse long-horizon workflows. Their progress remains constrained by expensive interactive experience, sparse outcome supervision, and growing visual histories. We introduce Qwen-CUA, a natively trained agent built on Qwen that observes only screenshots and acts through keyboard and mouse events, without DOM trees, accessibility metadata, or task-specific APIs. A lightweight history mechanism keeps recent visual evidence while compacting older context in chunks, controlling inference cost and preserving stable prefixes for cache reuse. To scale experience, we build a cloud rollout fleet and a broad collection of controllable web and desktop environments, synthesize state-grounded tasks with executable verifiers, and collect personalized long-horizon expert trajectories. These sources support a closed-loop training recipe that alternates supervised fine-tuning, reinforcement learning, large-scale rollout, outcome verification, and trajectory filtering. Rollouts are retained when they achieve verifiable outcomes and exhibit grounded actions, coherent progress tracking, and explicit state checking, reducing repetitive or unproductive behavior in subsequent training rounds. We evaluate Qwen-CUA under a pure computer-use setting across a broad suite covering general desktop control, long-horizon workflows, personalized and professional tasks, web interaction, and adversarial robustness. Qwen-CUA improves consistently over Qwen3.7, achieves 86.2 on OSWorld-Verified, and remains competitive with leading proprietary models. Further analyses show that its gains do not simply come from more verbose test-time reasoning and that robustness improves without sacrificing task utility. Finally, an internal browser deployment and hybrid command-line experiments expose the model to naturally occurring workflows and show how native interaction can complement specialized tools. Together, these results suggest that scalable verifiable interaction, careful long-context management, and iterative self-improvement provide a practical foundation for general-purpose computer-use agents.

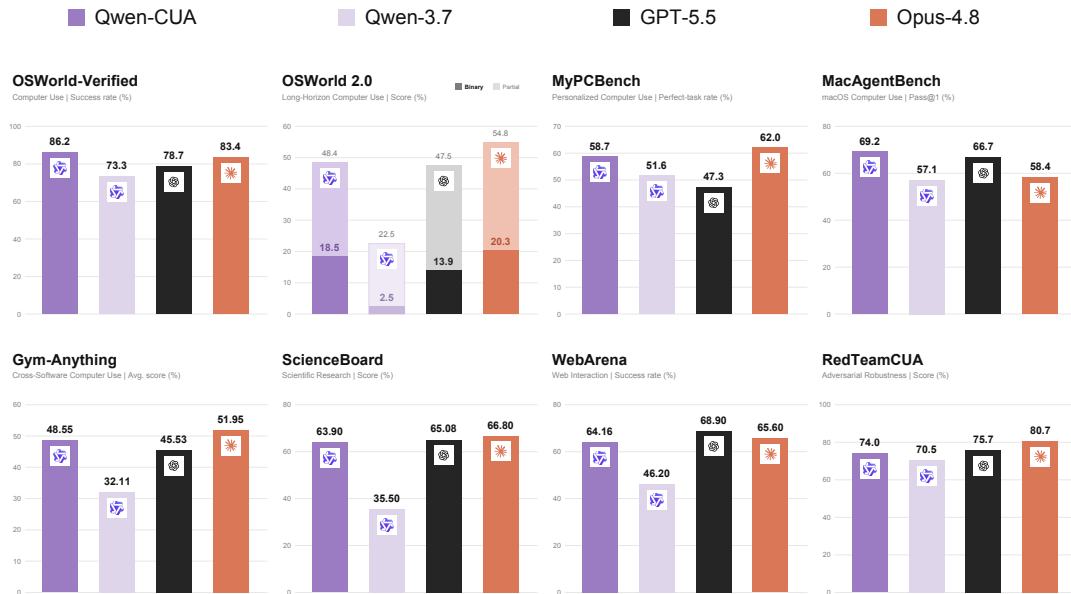


Figure 1: Qwen-CUA compared with Qwen-3.7, GPT-5.5, and Opus-4.8 across eight computer-use benchmarks. Qwen-CUA consistently improves over Qwen-3.7 and remains competitive with frontier models. OSWorld 2.0 reports binary completion (dark) and partial completion (light); all other panels show a single score. Higher is better.